8 Read the following article and then answer the questions that follow.

Analysis of braking of a vehicle

Modern vehicles are provided with different types of brakes. In cars, the foot brake is the most important in respect of control and safety of a vehicle while the hand brake is used as a reserve brake.

One of the principal braking parameters of a vehicle is the deceleration a. This criterion should satisfy the following condition

$$a \geq [\,0.10 + 0.85\,(\varphi - 0.20)]\,g$$

where φ is the coefficient of cohesion and g is the acceleration due to gravity.

The values of the deceleration of braking calculated according to the formula depend on the physical features of the wheel and road friction.

In reference books for experts' examination of traffic accidents as well as in scientific references on dynamics of vehicles, the provided value of coefficient of cohesion of tyres with dry asphalt is $\varphi = 0.80$. The value of the coefficient of cohesion equal to 0.80 may be applicable only to old cars and tyres produced about the year 1980. For the present-day cars, the maximum coefficient of cohesion φ is between 1.00 to 1.20, if braking takes place on dry asphalt.

The anti-lock brake system (ABS) is required to ensure distribution of braking forces between the wheels to prevent the wheels from locking and therefore causing the car to skid. The majority of modern vehicles are equipped with ABS, and their real braking distance is close to the theoretically calculated one based on the maximum values of the coefficient of cohesion. So, the deceleration of such vehicles may be close to $g = 9.81\ \text{m s}^{-2}$. For vehicles without ABS, the deceleration will be smaller.

(a) **(i)** Calculate the minimum deceleration of present-day cars with ABS while braking on dry asphalt.

minimum deceleration = …….. m s^{-2} [1]

(ii) Assuming a car experiences constant deceleration during braking, calculate the maximum braking distance for a present-day car travelling at 50 km/h.

braking distance = …….. m [2]

(iii) State and explain whether the braking distance of a vehicle will change while braking on wet asphalt.

……

…… [1]

(iv) Suggest two disadvantages of modern vehicles that have very large deceleration when braking if travelling on an expressway.

……

……

……

…… [2]

(b) A car has two important braking devices and they function independently. The foot brake is operated by foot and the hand brake is operated by hand.

When a car has a brake test, two sets of measurements are made:

1. The maximum braking force on the wheels produced by operating the foot brake.
2. The maximum braking force produced by operating the hand brake.

Typical data for a car of mass 900 kg are as follows.

Description	Maximum braking force / N
Foot brake	6700
Hand brake	2000

In order to determine whether or not the brakes are satisfactory, the data are applied to a chart (called a nomogram) like the one shown in Fig. 8.1. This chart has three vertical lines marked with scales.

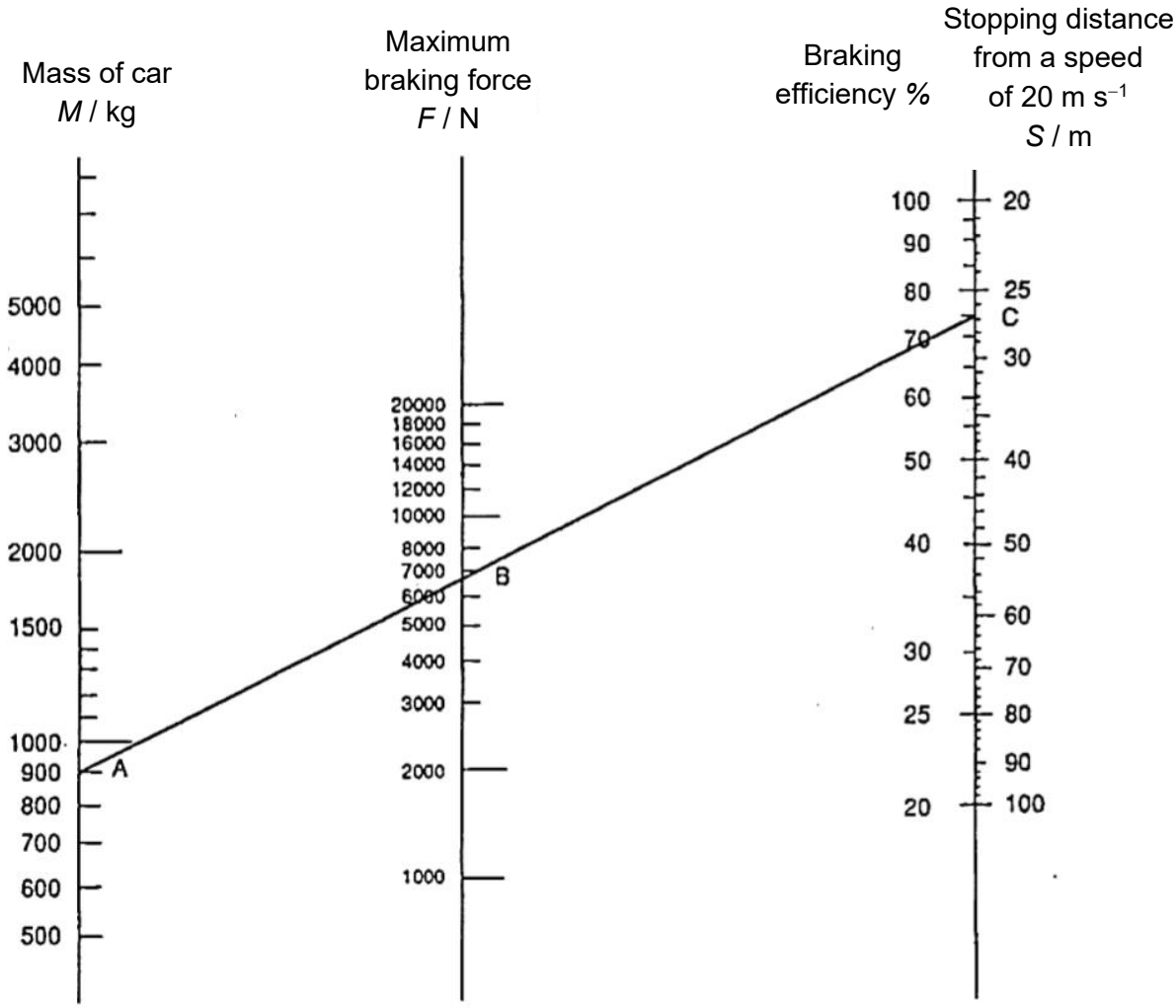

Fig. 8.1

The central vertical line is for the maximum braking force.
The left line is for the mass of the car.
The right line is for the braking efficiency and also for the stopping distance from an initial speed of 20 m s^{-1}.

The braking efficiency E is defined by the equation

$$E = \frac{\text{deceleration of car}}{\text{acceleration of free fall}} \times 100$$

As an example of the use of this chart for the car of mass 900 kg, the figures in the table show a maximum braking force for the foot brake of 6700 N. The point A corresponding to the mass and the point B corresponding to the braking force are joined to give a straight sloping line. This line is extended to cut the braking efficiency scale at the point C, and shows that in this particular case the stopping distance S from a speed of 20 m s^{-1} is about 27 m.

(i) In Fig. 8.1, draw a line to represent the results of the hand brake test on the car of mass 900 kg.

(ii) Read from the chart the braking efficiency corresponding to hand brake test.

braking efficiency = …….. % [1]

(iii) Calculate the deceleration corresponding to this value of braking efficiency.

deceleration = …….. m s^{-2} [1]

(iv) Show that the deceleration is approximately g = 9.7 m s^{-2} when both foot brake and hand brake are applied to stop a 900 kg car of speed of 20 m s^{-1}.

[1]

(v) Suggest why it is not advisable to use the braking technique in **(b)(iv)**.

…….

……. [1]

(c) Now consider a car of mass 1300 kg. The data for maximum braking force and stopping distance from 20 m s^{-1} are given in Fig. 8.2.

F / N	*s* / m
3000	88.0
4000	66.0
5000	54.0
6000	45.0
7000	
8000	32.5
10000	27.0
12000	23.0

Fig. 8.2

(i) Use Fig. 8.1 to determine the stopping distance when *F* = 7000 N.
Show your construction on Fig. 8.1.

stopping distance = …….. m [1]

(ii) On Fig. 8.3, plot the point corresponding to *F* = 7000 N and draw the line of best fit for all the points. [1]

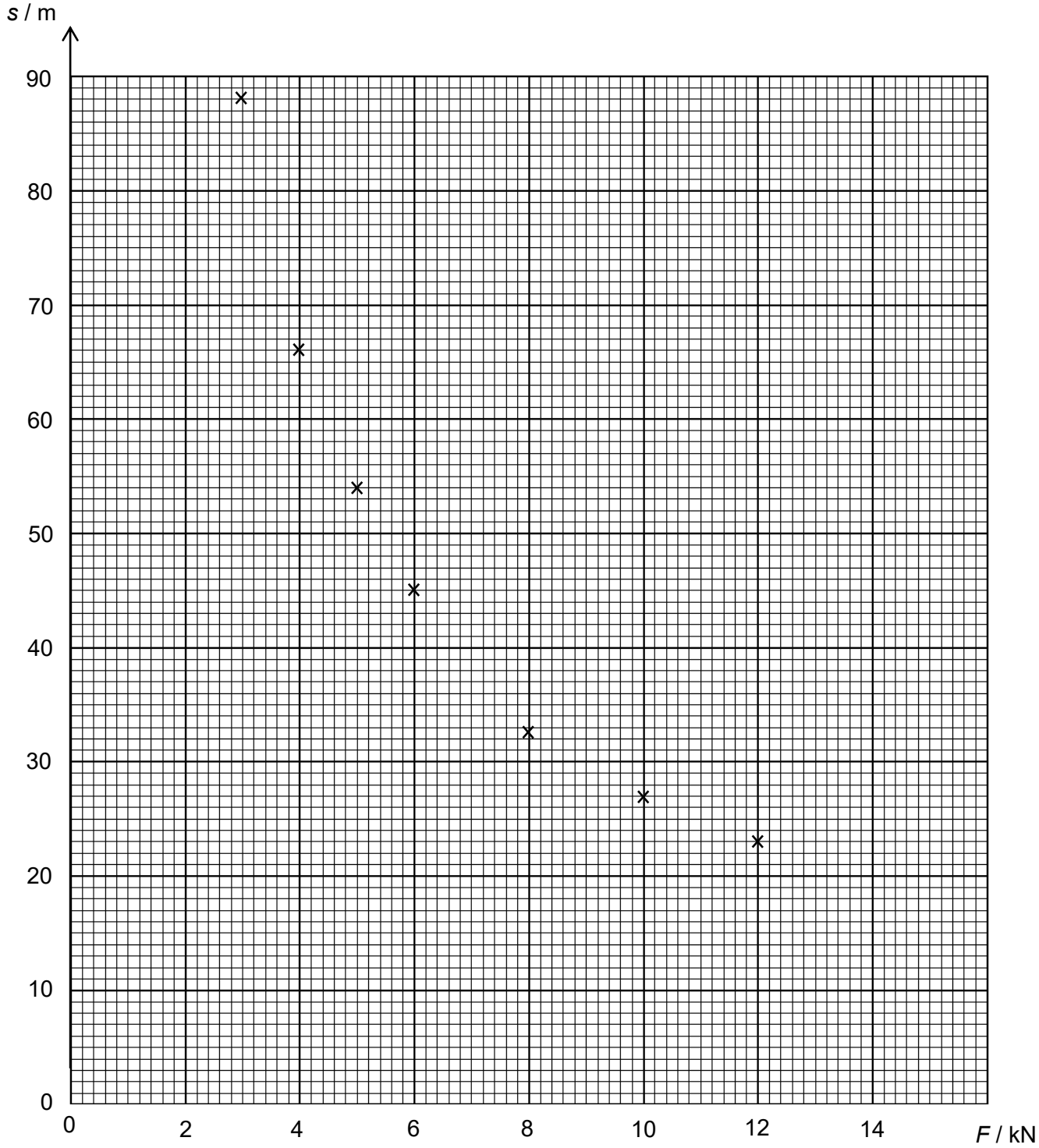

Fig. 8.3

(iii) Use Fig. 8.3 to determine the work done when the maximum braking force is 9000 N. State any assumptions made in your calculations.

work done = …….. J

…….

……. [2]

(d) **(i)** A passenger of mass 70 kg was not wearing a seat belt and was sleeping at the front seat of a car. The car of mass 1300 kg was travelling at 20 m s^{-1} when a maximum braking force of 12000 N was applied on it.

Using Fig. 8.1 and the equation for braking efficiency, determine the minimum force that the passenger would have to apply to prevent himself from flying off the seat. Show your working clearly.

force = …….. N [3]

(ii) Suggest whether the passenger would be able to prevent himself from flying off the seat.

…….

…….

……. [1]